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Research on false review detection Methods: A state-of-the-art review

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ABSTRACT

Fake reviews are popular today where product reviewers write the reviews without experiencing or purchasing the product on e-commerce and restaurant portals. Currently, the false review recognition method uses the systematic review process to extract, summarise, and classify the meaningful content of the research, compare and analyse the representation power of various false attributes, and the recognition method's performance. Feature design and recognition method design are the key steps for false review text recognition. The procurement of a large-scale labelled review dataset is difficult in recent research. They were only identifying fake review texts used as the core of the discussion. The article presents an assessment of fake reviews detection in different domains (hotels and e-commerce). In this article, we have also identified the relation between fake reviewers and groups of fake reviewers. We have analysed and pointed out the existing research problems in data acquisition, false feature design, and recognition method design to suggest future research on false review detection.

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1. Introduction

The development of Web 2.0 has given the freedom to Internet users to express their opinions on social media and web portals. Users share experiences by posting their opinion on a product, service and complaint on social media and e-commerce platforms. As Internet users and online web portals are increasing, the number of product, service and complaint reviews is increasing very rapidly in proportion to it. According to the American review site Yelp (Alamoudi and Azwari, 2021), as of the third quarter of 2020, the site had more than 220 million user reviews shown in Fig. 1, and the distribution of reviews on different businesses basis shown Fig. 2 (Ching, 2019). Product reviews have a significant impact on consumers' purchasing decisions (Samha and Xia, 2008; Samha et al., 2014). Some companies hire spammers who write false product reviews to promote their products or demote competitor products.

Jindal et al. (Jindal and Liu, 2007) first proposed the concept of "Review Spam" in 2007. They divided spam reviews into three categories: untruthful reviews, reviews on brands only and non-reviews shown in Fig. 3. Such reviews are widely found in various online shopping and review websites (Ott et al., 2012) and have the characteristics of great influence, strong concealment, and high identification difficulty, making it a focus of attention in related research. There are three main types of targets for the research and identification of spam reviews: false reviews, false reviewers and groups of fake reviewers.

Literature (Rastogi and Mehrotra, 2017) published on the subject of false review identification, of which 63% was the target of false review text identification. The further author statistics found that this proportion continues to rise, as of 2017 reached 66% annually. The recognition of false review text is a hot topic in this field and constitutes research on false reviewers and group identification methods. However, there is still a lack of literature reviews focusing on the research status of false review text recognition methods in research, especially the lack of review and comparison of key issues such as datasets, features, models and research methods. Therefore, this article refers to Barbara Kitchenham's (Kitchenham, 2004) systematic literature review process to collect, screen, record, and analyse the relevant literature on spam review detection. The main problems addressed by this research include:

1. To identify the recent trend of spam review detection methods and the types and characteristics of data, features, models used in each method.
2. To compare the contribution of different types of features and methods to improve prediction performance.
3. To reveal the problems existing in the current research and propose future research directions.

2. Literature collection

We present the literature data procedure to obtain research articles on the fake review detection area, as shown in Fig. 4. We used "Review Spam Detection", "Fake Opinion Analysis", "Deceptive Reviews Detection", "Opinion Spam Detection", "Fake Reviews Detection", "Spam Review Detection", "Review Spammer", and "Social Media Spam Detection" as the search terms for subject search.

The search scope set to SCI journals or High index conference research papers published between 2007 and 2020. Literature search performed into different databases such as Web of Science, ACM Digital Library, IEEE Xplore, ScienceDirect, ACL, and LNCS, Cornell University, SSRN. More than 900 articles related to the subject during the initial search. Quality assessment and reference tracing have shown 322 research papers in Fig. 5 and the publication distribution in Fig. 6 that meet our quality requirements.

Out of which 162 research journal articles were identified with false review texts, 11 old review papers and 139 articles from useful conferences. The number of documents excluded under the three criteria is the title, judgment methods (Read literature, keywords, Deep reading) and references criteria used in the literature quality assessment.

According to the systematic literature review (Kitchenham, 2004), the final 112 research papers were extracted when the research articles read one by one. Based on the classification and analysis of the extracted data, the existing technical research route on spam review detection can be divided into steps shown in Fig. 7: review dataset acquisition, data pre-processing, feature engineering, model building, outcome evaluation.

The review dataset mainly comprises business reviews and crowdsourcing reviews; spam review features are mainly divided into categories: review-based features, reviewer dimension features, and relationship dimension feature, as given in Fig. 8. Machine learning relies on label data which is limited, and it takes a lot of time and resources to label the data. Existing research mainly uses evaluation methods and indicators for text classification and information retrieval tasks in terms of results evaluation. In the above-given steps, spam review features and the design of recognition methods are the key issues in this field.

Finally, 112 research papers with the most representative features, methods and results were selected to review the key issues of the research on fake review detection, summarise and compare the characteristics, performance and applicability of existing recognition methods. Nowadays, the identification of fake reviews is not limited to the institute label research stage, but the major e-commerce websites are getting attention and practice. For example, Amazon, Yelp, Taobao and Dianping already introduced the

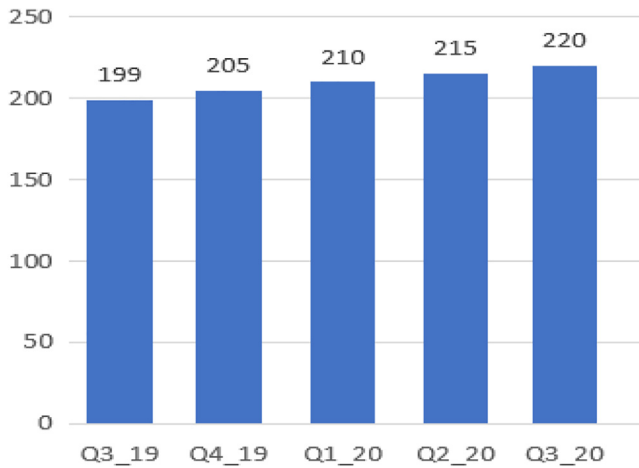


Fig. 1. Reviews contribution from 2019 to 2020 on the Yelp website (Ching, 2019).

rules for identifying and punishing false posting, public review network, and Yelp implemented a classification display mechanism for reviews (Li and Chen, 2014; Ott et al., 2011). However, because commercial websites have different methods for identifying false reviews, technology is a trade secret and is not disclosed to the public. According to the information we gather, these websites' feature data dimensions are still mainly composed of review content, reviewer behaviour, and interrelationships consistent with the feature types summarised in this article. They may also include, for example, user operation logs, registration information, the internal data of the website that cannot be publicly obtained, such as mouse click behaviour and Internet Protocol address (Mukherjee et al., 2013).

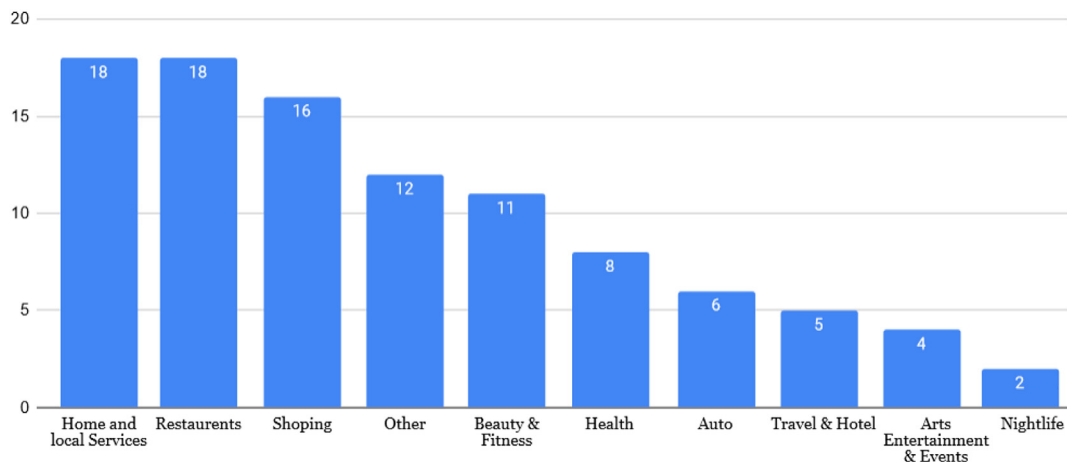


Fig. 2. Business-based review distribution on Yelp (Ching, 2019).

1 Negative deceptive Review

I stayed at Sofitel with my husband for a weekend, and I will NEVER be staying again! The staff was rude, the room was dirty, and the decor is ugly and garish. AVOID at all costs!

2 Positive deceptive Review

Great hotel! Went to see the Museum of Contemporary Art, which was great, but this hotel almost had it beat! The rooms (and even halls) are very beautifully done. Great service too.

Fig. 3. Example of a review (Ott et al., 2012).

3. Spam review dataset

According to their sources, the review datasets used in existing research can be divided into two types: business review data and crowdsourcing review data. Real-world shopping websites, restaurants website and review portals are the sources of business review data. This type of data is usually unlabelled, and false and non-fake reviews are mixed. Research on business review data application mainly adopts rule labelling, manual labelling, and commercial labelling to realise data labelling. Crowdsourced review data includes two parts: real and fake review data. Among them, real reviews are generally obtained by researchers from commercial websites based on certain criteria.

In contrast, fake reviews are posted on crowdsourcing platforms, and participants are paid on a fake review basis with accurate annotations. Thus, it takes much time to annotate the review data. The characteristics of the two types of review data and different labelling methods are shown in Table 1. Table 2 shows the representative public reviews datasets, and their acquisition methods are listed.

4. Feature engineering

Feature Design

Fake review feature design extracts and constructs feature from the review dataset that can effectively characterise fake or non-fake reviews and suspicious users via fraudulent comment activities. According to the different feature analysis dimensions, the following types of fake review features are extracted from reviews: review features, reviewer features, product feature, seller feature and relationship features shown in Fig. 9.

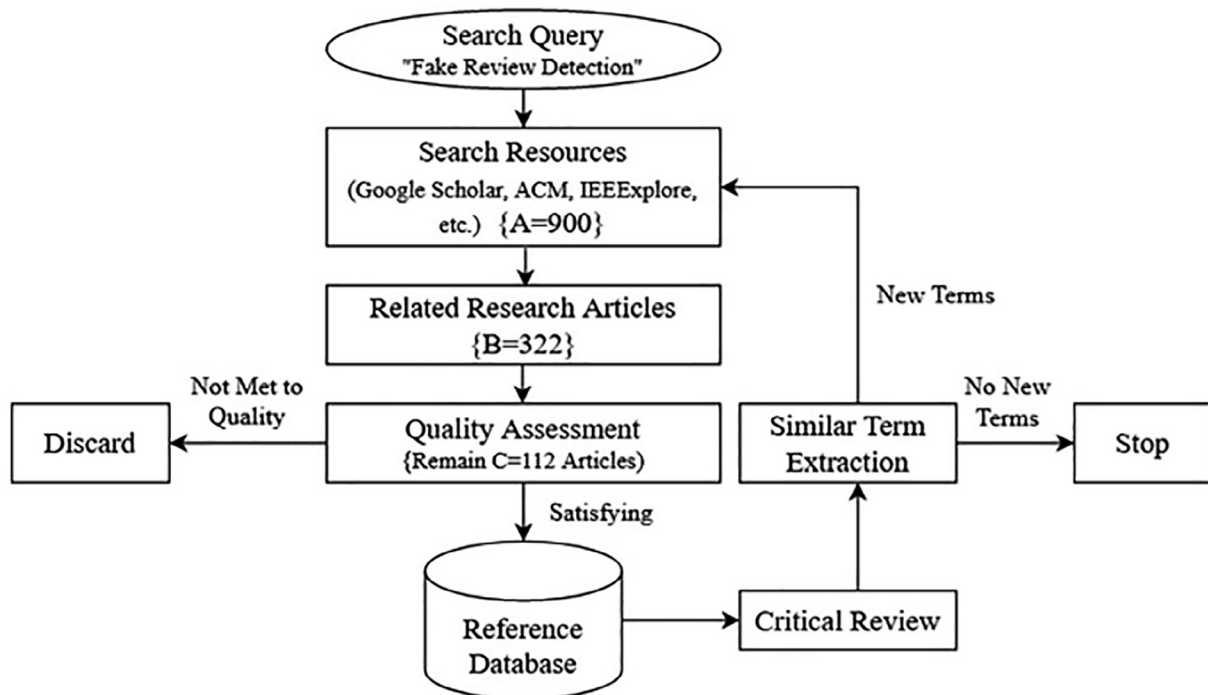


Fig. 4. Literature collection Process.

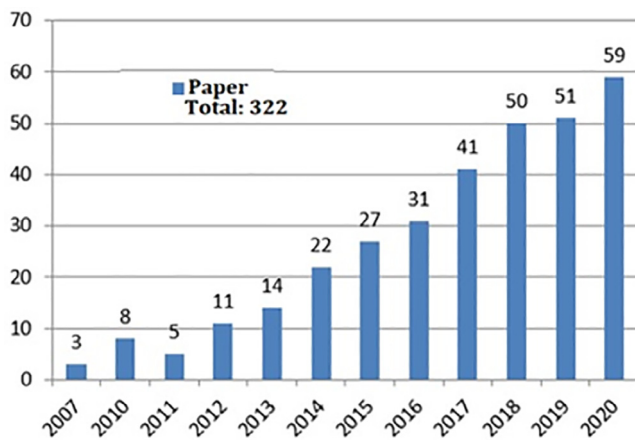


Fig. 5. Research papers after the first selection criteria from 2007 to 2020.

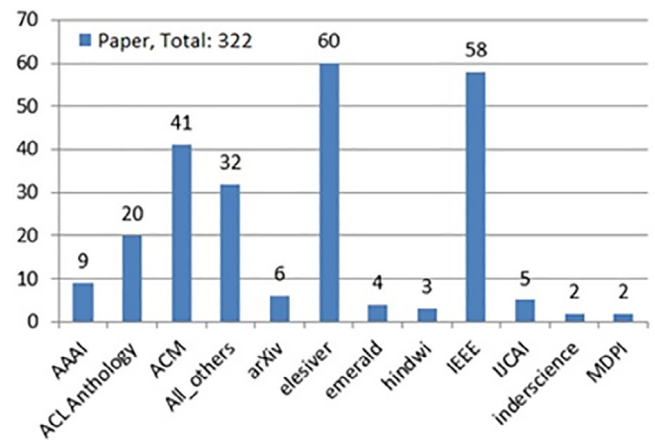


Fig. 6. Research papers found on publication after the first selection criteria from 2007 to 2020.

4.1. Review dimension feature design

Reviewers express their feelings, opinions, or emotions about the review object through language. Fake reviews are lies fabricated by the reviewers. Their content is based on unreal experiences and leaves false clues in the language use of reviews. The feature design of the review dimension is centred on the text of the review, and lexical, syntactic and semantic analysis is performed on the vocabulary, sentence, text and semantics of the comment to obtain its linguistic and text features. The review dimensions feature is shown in Fig. 10 and Table 3 of their extraction methods.

4.2. Reviewer dimension feature design

A review is only related to one reviewer and one reviewed product, and the false clues that can be derived from the feature

design of the review dimension are very limited. The reviewers view that it can post multiple reviews to multiple products and leave more false clues. Fake reviewers achieve their goals, and their review writing behaviour is quite different from that of normal reviewers. Therefore, the dimensional feature design of the reviewer is centred on the behaviour of the reviewer. Mining and design can effectively characterise the behaviour differences between false and real reviewers. The realisation process mainly has three steps:

1. Observe and analyse the abnormal behaviour of the reviewer.
2. Summarise and design characteristics that describe abnormal behaviour.
3. Establish a mathematical model to quantify the characteristic value.

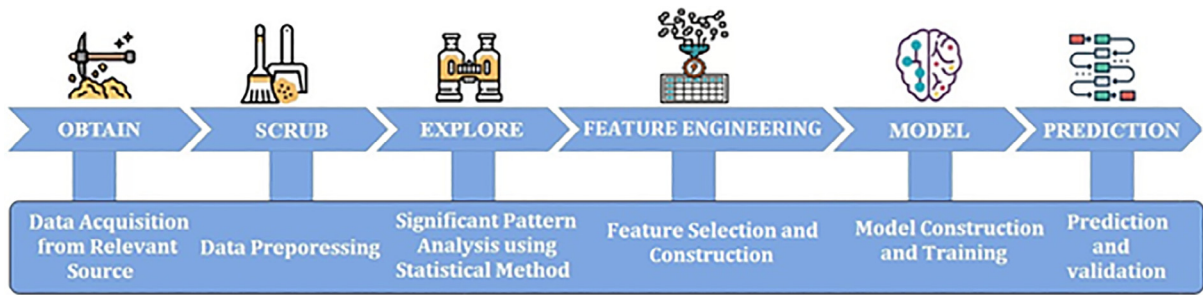


Fig. 7. General process of SRD (Mason Chris Wiggins 2010).

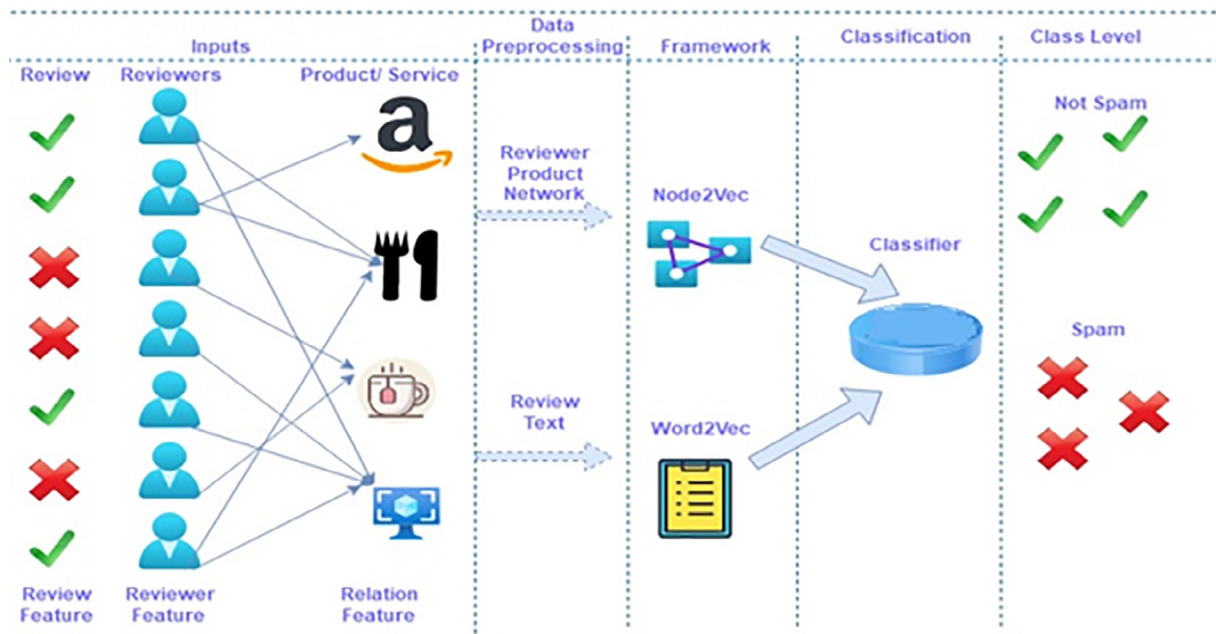


Fig. 8. The general process of review spam detection.

Table 1
Data labelling methods and representative public datasets.

Dataset types	Labelling method	Advantages	Disadvantages	Dataset description
Business review data	Rule-Based: - marked review as spam if the text-similarity $\geq 90\%$ (Jindal and Liu, 2008) Manual labelling: Invite Turker to perform manual labelling based on the judgment criteria of false Reviews (Li et al., 2011) Commercial annotation: commercial websites use their internal algorithm to classify and display reviews. For example, Yelp divides reviews into recommended reviews and non-recommended reviews (Mukherjee et al., 2013)	Simple and efficient, large label There is an absolute basis for judgment High accuracy, large label	Effectiveness is difficult to determine (Mukherjee et al., 2013) Low accuracy rate (Li and Chen, 2014); a small amount of labelling, high labelling cost (Li et al., 2011) The recall rate is difficult to determine (Li and Chen, 2014; Li et al., 2014)	5.8 million Amazon review data 64 K, Yelp reviews data, multidomain Yelp review data, Dianping small dataset
Crowdsourcing review data	Crowdsourced annotation: artificially screen real reviews from e-commerce websites, and artificially falsify fake reviews through crowdsourcing platforms such as AMT (Ott et al., 1107; Li et al., 2014; Ott et al., 2013)	High accuracy rate, can reflect the psychological characteristics of reviewers (Ott et al., 1107)	A small amount of labelling, labelling high cost, a small amount of information, data distribution does not match with the real-world (Mukherjee et al., 2013)	“gold standard” hotel review dataset, “gold standard” multidomain review dataset

4.2.1. Duplicate review behaviour
The text similarity calculation method is relatively simple, and the principle is easy to understand. It is to calculate the ratio of the intersection and union between word sets. The larger the value, the more similar the two characters. When it comes to massively parallel computing, this method has certain advantages inefficiency.

To reduce the time cost of writing false reviews, spammers duplicate the reviews of themselves or others, resulting in greater similarity in the content of the reviews (Ee-Peng Lim et al., 2010; Mukherjee et al., 2012). The cosine content similarity technique can apply to find duplicate reviews (Mukherjee et al., 2013) written by the reviewer.

Table 2
Different public reviews datasets used in literature.

Annotation method	Dataset name	Dataset description	Dataset fields
Rule-based	Amazon product review	5.8 M reviews and 2.15 M reviewers	Books, DVD, music, products (Jindal and Liu, 2008)
	Amazon book review	6.8 K reviews and 4.8 K reviewers	Books (Formaciari and Poesio, 2014)
	TripAdvisor booking Review	2.8 K reviews	Hotels (Hammad, 2013)
Human-Based	Epinion	6 K reviews	Products (Li et al., 2011)
	TripAdvisor	3 K Reviews	Hotels (Ren et al., 2014)
Algorithm-Based Filtering	Yelp hotel	67.4 K reviews and 38 K reviewers	Hotels and Restaurants (Mukherjee et al., 2013)
	Yelp hotel	3.59 M reviews and 16 K reviewers	Hotels and Restaurants (Rayana and Akoglu, 2015)
	Yelp hotel	6.086 M reviews and 2.6 M reviewers	Hotels and Restaurants (Rayana and Akoglu, 2015)
	Dianping	10 K reviews and 9 K reviewers	Restaurants (Li and Chen, 2014)
Amazon Turker	TripAdvisor Hotel Review	800 reviews	Hotel (Ott et al., 1107)
	TripAdvisor Hotel Review	1600 reviews	Hotel (Ott et al., 2013)
	TripAdvisor multidomain	3032 reviews	Hotels, Restaurants, Doctors (Li et al., 2014)

4.2.2. Abnormal scoring behaviour

Spammers can achieve the purpose of increasing or decreasing the comprehensive scores of review products by giving scores that are inconsistent with normal reviewers or even extreme (Jindal and Liu, 2008).

1. Extreme Rating:- Determine whether reviewer $\times$ always gives an extremely positive or negative score to the specific product p from the 5-star rating system to review $R1$. Rating 1-star is extremely negative, and 5-stars is extreme positive (Mukherjee et al., 2013).
2. Percentage of Positive Reviews:- Calculate the degree to which the reviewer tends to post positive reviews. Among the 5-star rating reviews with greater than 4 stars are positive reviews (Mukherjee et al., 2013).

3. Rating Deviation:- Calculate the rating of a certain product by the reviewer and the degree of inconsistency between the reviewer and the average rating. The denominator indicates the maximum rating deviation of 4 out of the 5-star rating system (Geli Fei et al., 2013).

4.2.3. Early review behaviour

Spammers tend to review the product when it is first launched (Ee-Peng Lim et al., 2010), which can influence consumers' impression of the product to the greatest extent.

1. Early Time Frame:- Calculate how early a reviewer reviews a product's latest review time and launch time. Here 8 months is the early review threshold, and the estimated value is 7 months (Mukherjee et al., 2013).

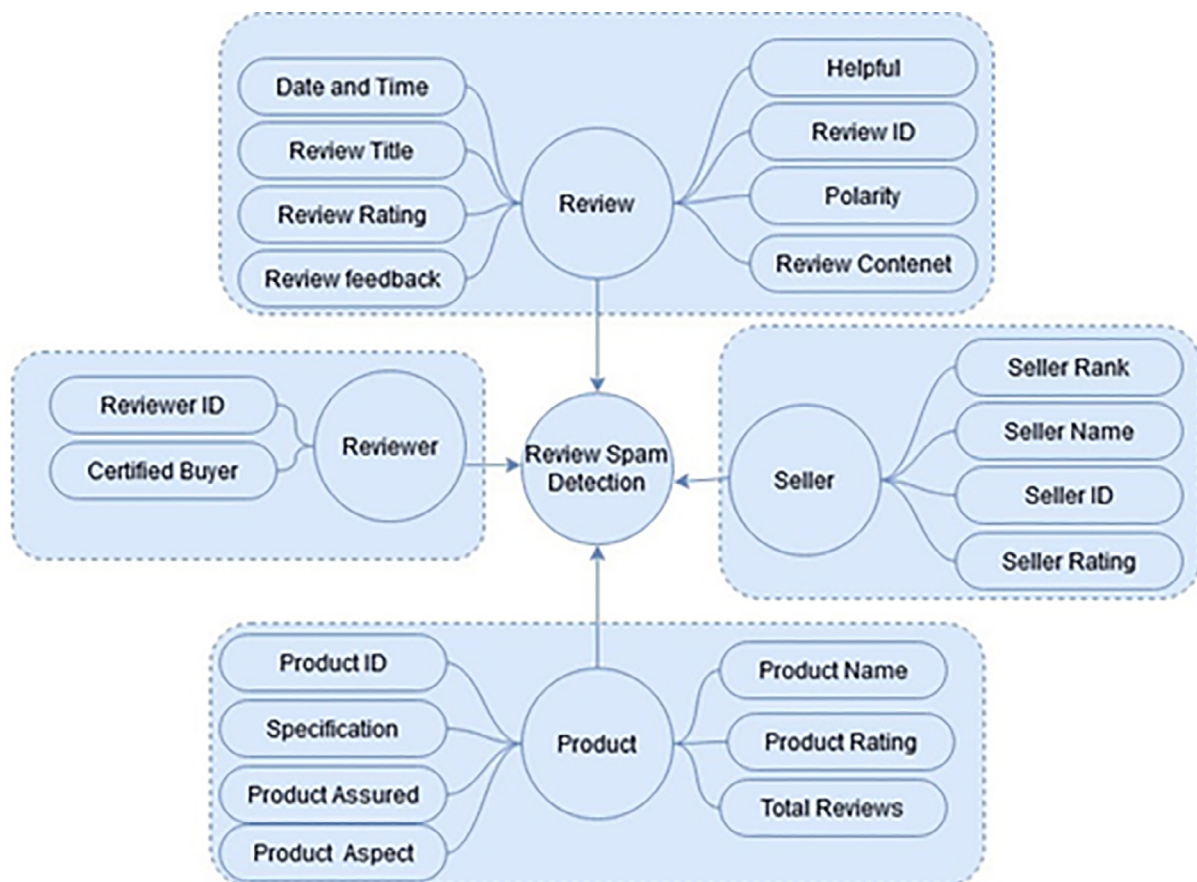


Fig. 9. Relationship of review, reviewer, product and seller with limited feature.

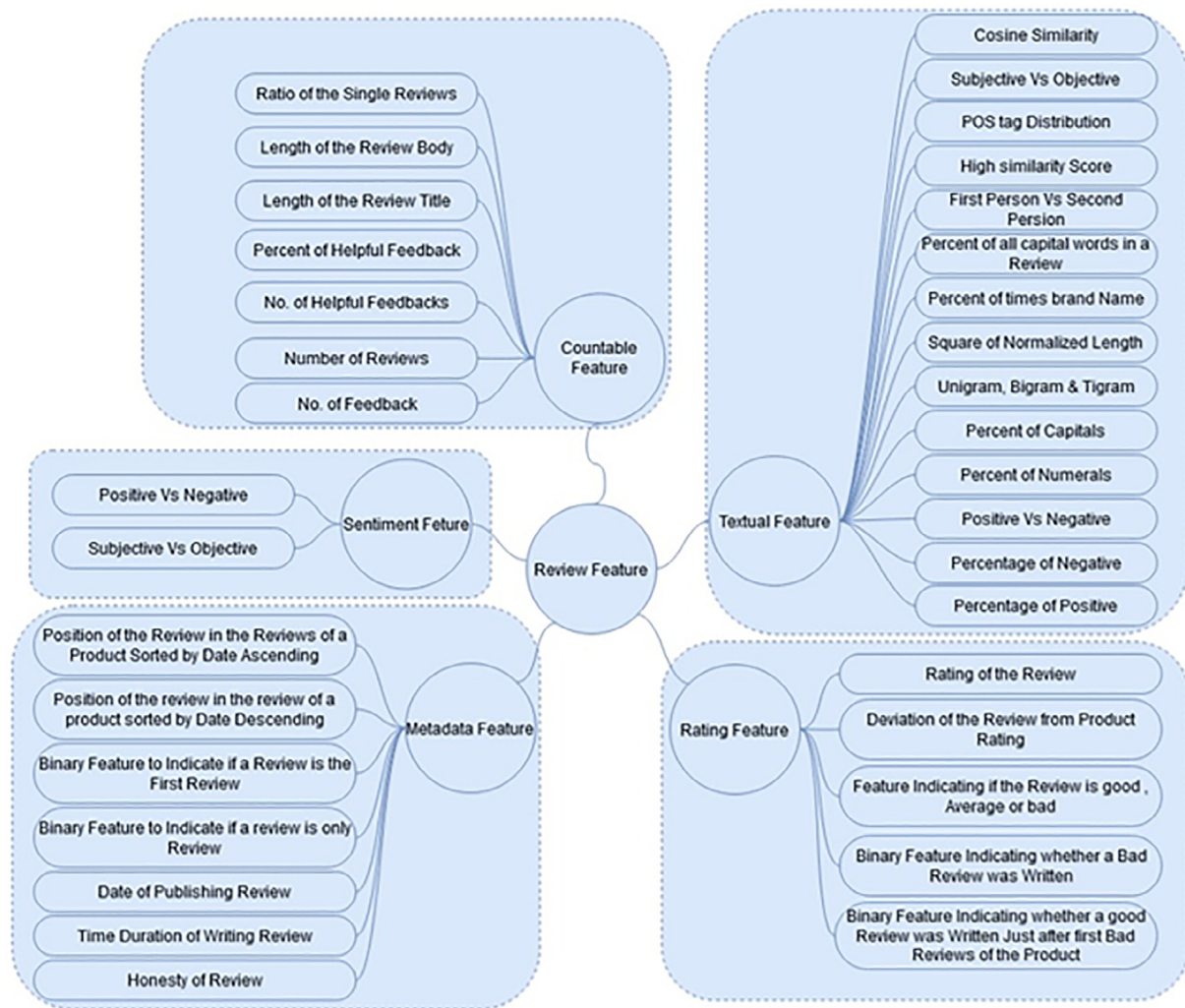


Fig. 10. Review content features.

Table 3

Features and extraction methods of common reviews.

Feature Category	Feature	Feature Function	Method of Extraction
Linguistic features	Bag of Words (1-gram, 2-gram, n-grams)	Express text in vector form and calculate the similarity of comment text	Lexical analysis, build a bag of words model or n-grams model, commonly used are Unigram, Bigrams and Trigrams.
	Part-of-Speech feature	Capture differences in the part-of-speech distribution of real and false reviews	Lexical analysis, perform part-of-speech tagging and word frequency statistics
	Linguistic Inquiry and Word Count Feature	Understanding false comment text from the perspective of linguistics and psychology	Lexical analysis, using the output of Linguistic Inquiry and Word Count tool
	Grammatical Structure	Get the deep syntactic structure of the review text	Syntactic analysis, using probabilistic context-free grammar or dependency analysis tree
	Emotion	Catching extreme emotional tendencies of false comments	Semantic analysis, sentiment analysis
Text feature	Textual representation	Get the deep semantic representation of reviews	Semantic analysis, through the establishment of word embedding model, deep learning model for text representation.
	Stylistic Style	Get the reviewer's writing style characteristics	Lexical and syntactic analysis, based on vocabulary and syntactic feature analysis and extraction
	Metadata	Obtain the attributes of reviews, reviewers, and review objects from a perspective other than the review language	Extract data related to reviews other than the review body, such as review length, review time, rating, ID, product ID.

2. The First Review ratio:- Calculate the proportion of all reviews of reviewers that are the headline reviews.

4.2.4. Sudden review behaviour

The fake reviewer made multiple reviews in a short time interval to achieve the purpose of quickly influencing the compre-

hensive score of the review object and manipulating the emotional tendency of the review (Mukherjee et al., 2012; Xie et al., 2012).

1. Reviewing Burstiness:- Calculate the closeness between the reviewer's first review time and the last review time. The

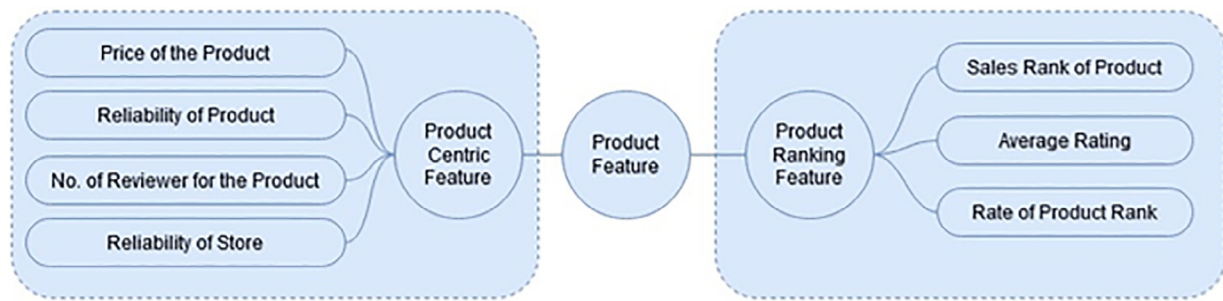


Fig. 11. Product based features.

threshold of the burst period and the estimated value is 20 days (Mukherjee et al., 2013).

2. **Burst Review Ratio:-** Calculate the proportion of all reviews by a reviewer that is sudden reviews. The collection of reviews in all reviews of the reviewer is a review burst period (Mukherjee et al., 2013).
3. **Singleton Ratio:-** A single review refers to a reviewer who has only posted one review, and the possibility of being a false reviewer is high (Xie et al., 2012). The collection of reviews of products (Ruan et al., 2020).

4.2.5. Frequent reviews

The time and number of reviews posted by normal reviewers are generally random. In contrast, the time and number of reviews by false reviewers have certain rules due to specific tasks, such as a certain number of reviews every day and every month (Mukherjee et al., 2013).

1. **Maximum Number of Reviews:-** Calculate the ratio of the maximum number of reviews posted by reviewers in a single day to the maximum number of reviews posted by other reviewers in a single day (Mukherjee et al., 2013).
2. **Posting-Frequency:-** Calculate the maximum difference between the actual number of reviews and the average number of reviews from the reviewer per month (Rout et al., 2017).

4.2.6. Targeted reviews

The review objects of normal reviewers are usually random, while the reviews posted by fake reviewers are highly targeted, such as only commenting on specific products, brands, or shops (Li et al., 2016).

1. **Brand Deviation Score:-** Calculate the distribution difference of the number of reviews on different brands by reviewers representing the brand, the probability of the brand being reviewed, measured by entropy (Li et al., 2011).
2. **The ratio of Outlier Products Reviewing:-** Outlier products refer to products reviewed by only one reviewer. This feature calculates the proportion of outlier product reviews among all reviews of the reviewer (Wang et al., 2020).

3. **Review Density:-** Calculate the density of reviews by reviewers on a specific product category, store, or time (Melleng et al., 2019; Li et al., 2016).

Review metadata is also an essential data source for behavioural feature analysis, for example, review length, purchase history, review time, and location. The review length of fake reviewers is usually shorter (Mukherjee et al., 2013); the credibility of reviews with real purchase records is higher than that of reviews without purchase records (Ji et al., 2020; Geli Fei et al., 2013; Rout et al., 2017). Authors (Li et al., 2015) analysed the metadata patterns, timing and spatial patterns of real and fake reviewers based on many commercial annotation data of Dianping website containing Internet protocol addresses. They proposed a series of timing features and spaces with strong false representation. Also, many metadata can reflect the reviewers' preferences and social behaviour characteristics, such as the reviewers' photos, check-in, favourites, and the behaviours of adding friends, following, replying, and liking between reviewers. Social relationships analysis helps to identify false reviews (Zhang et al., 2016; Rahman et al., 2015; Viviani and Pasi, 2017; Goswami et al., 2017). Literature (Mukherjee et al., 2013; Zhang et al., 2016) pointed out that the reviewer behavioural feature has better false representation power than the review's dimensional feature. However, this method of extracting features based on the observation and analysis of the reviewer's abnormal behaviour still has shortcomings.

1. It is impossible to find fake reviewers whose behaviour is not much different from normal reviewers (Chengzhang and Kang, 2015).
2. Once the fake reviewers realise this abnormal behaviour-based detection mechanism, they quickly adjust their behaviour patterns to eliminate the difference from normal reviewers (Xue et al., 2019).
3. Features based on manual observation are subjective.

Some observed behavioural clues cannot fully prove that the reviewer must be a spammer. For example, the suspicious behaviour of posting multiple reviews on the same product may also

Table 4

Graph model of relationship dimension features.

Types of graph	Representative model	
Bipartite graph model	The node has no additional features in the reviewer-product graph model (Fang et al., 2020; Akoglu et al., 2013).	The nodes in the reviewer-review the graph model is enclosed with related reviewers and review features (Yuqing et al., 2013; Shehnepoor et al., 2017; Saumya and Prakash Singh, 2020).
Tripartite graph model	In the reviewer-review-product graph model, each node has reviewers, reviews, and product features (Rayana and Akoglu, 2015).	reviewer-IP-review graph Model, each node has a corresponding reviewer and review features, and review nodes that share the same IP or reviewer nodes tend to have the same label (Ren et al., 2014)

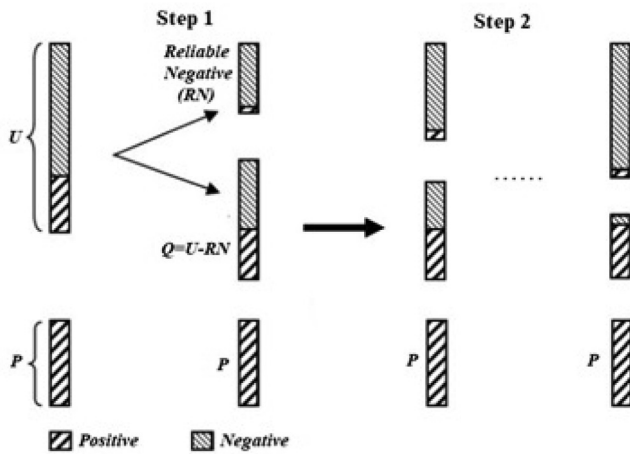


Fig. 12. Positive-Unlabelled Method Based on Two-step.

be caused by normal reviewers buying the product multiple times (Guo et al., 2021).

4.3. Interrelation dimension feature design

It is relatively easy for fake reviewers to imitate the language or behaviour patterns of real reviewers. Still, it is difficult for them to imitate network relationships between different subjects (reviewers, reviews, and product relationship shown in Fig. 7 and product feature shown in Fig. 11) (Zhang et al., 2020; Mengxiao et al., 2019). The dimensional feature design of the relationship is centred on the relationship between the subjects, and the relationship between the subjects is analysed as a false feature. According to literature research, most of the existing researches are based on graph theory to establish a graph model of the relationship between subjects and reveal the false behaviour of the reviewer through the analysis of the adjacency relationship of the nodes in the graph and the characteristics of the additional attributes of the nodes (Xue et al., 2019). According to the different number of node types in the graph model, the graph model used in the spam review detection can be divided into two types: bipartite graph model and tripartite graph model, as shown in Table 4. In research, the correlation feature is more used in the identification of spammer groups. Fig. 12.

5. Methodology design

The methodology design aims to build a spam review detection model based on various review features and make a final prediction on the review text label. Because machine learning depends on label data, it can be divided into methods using labelled data, semi labelled data, and unlabelled data.

5.1. Labelled Data-Based detection method

The labelled data-based Detection method is the most widely used method in existing research. It is mainly based on the text classification method of supervised machine learning to divide reviews into two classes, fake and non-fake. The core issues of the method are data set construction, feature selection and classifier construction. The training and testing datasets used by the existing classification methods are mainly crowdsourced and commercial labelled data. Commonly used classifiers include Support Vector Machine, Naive Bayes, Logistics Regression, Decision Tree. According to the types of review features used in the construction

of the classification model, the existing classification methods can be divided into classification methods based on the dimensional characteristics of reviews and classification methods that combine the reviewer's characteristics and their relationship.

5.1.1. Classifier based on review features

Early research mainly extracted review features by analysing the review text's vocabulary and sentence levels (Ott et al., 2011; Li et al., 2014; Sadman et al., 2020; Feng et al., 2012; Hussain et al., 2020; Qiongkai and Zhao, 2012). At the linguistic level, ott et al. (Rastogi and Mehrotra, 2017) worked on n-gram, and part of speech features, the validation accuracy of the Support Vector Machine classifier on the "Gold Standard" dataset was 90 % approx. At the sentence level, Authors (Feng et al., 2012; Qiongkai and Zhao, 2012) passed the features of Unigram and PCFG (Probabilistic Context-Free Grammars) together, with the accuracy of the model exceeding 90 %; There are studies based on lexical and syntactic features to analyse reviews from the perspective of text writing style (Hussain et al., 2020; Shojaei et al., 2013; Dewang and Singh, 2015). For example, Dewang (Dewang and Singh, 2015) proposed feature sets that describe the linguistic style, such as lexical density, symbol ratio, average phrase length. They used the Support Vector Classifier on a "gold standard" dataset that gained a 91.51 % F-measure value. Authors (Abri et al., 2020; Li et al., 2014) proved that LIWC features significantly affect the improvement in classification accuracy. Karami et al. (Karami and Zhou, 2015) proposed falseness indicators based on LIWC features, such as punctuation scores, positive or negative sentiment scores, differences between personal interest related words and general vocabulary scores. The recognition accuracy rate on the "gold standard" dataset exceeds 93 %. In order to dig deeper into the false clues at the semantic level of reviews, some studies have applied word embedding models (Kumar, 2003; Zhang et al., 2016; Cao et al., 2020; Hernandez-Castaneda and Calvo, 2017) and neural network models (Hajek et al., 2020; Anass et al., 2020; Fahfouh et al., 2020; Ren and Ji, 2017; Li et al., 2017; Sun et al., 2016) to obtain the text representation characteristics of reviews, which is shown in the problem of handling reviews data domain migration good performance.

However, the accuracy rate of classification models based on the review feature is relatively high, especially since crowdsourced data's accuracy rate is close to 90%. But Mukherjee (Mukherjee et al., 2013) reported that such models' accuracy on annotated datasets is 68%, much lower than the crowdsourced dataset results. The reason is that the sources of real and false reviews in the crowd-sourced dataset are different, the language of the two is also quite different, and the data distribution does not match the real-world distribution. Other studies (Hussain et al., 2020) tested the classification model based on reviews' content features on a manually labelled dataset. The classification accuracy rate was 84%, and the model was not stable. The reason is that different artificially annotated datasets have large differences in the review field, language, data volume, and annotation quality, which affects the performance of the model. Feature selection and optimisation improved the performance of the machine learning algorithm. Author Laith Abualigah (Abualigah et al., 2021; Abualigah et al., 2021; Abualigah and Dulaimi, 2021; Abualigah and Diabat, 2021) survey and proposed different optimisation algorithms are SCAGA, Aquila optimiser, and arithmetic optimisation algorithm, which can apply in review spam detection. Aquila Optimizer worked in four ways. It Selecting a search location by flying high along with vertical, exploiting within a convergence, searching within a different search space by contour flight with short glide attack search space by flying low with slow landing attack and swoop to walk and catch prey. The arithmetic Optimisation Algorithm utilised four basic mathematical operations are X, /, + and -. Both algorithms optimal values compared to x7 for

Aquila 5.2994 on optimal weight 3007.74, arithmetic algorithm 5.2867 on optimal value 2997.92 and SCAGA algorithm prevent local minima problem compared to SCA algorithm.

5.1.2. Classifier based on reviewers and relationship features

Most researchers choose commercial annotation comments to construct experimental datasets because of crowdsourced annotation and manual annotation of data. It also integrates reviewers and relationship dimensional features to improve the classification performance in establishing classification models and using review features. Mukherjee et al. (Mukherjee et al., 2013) added reviewer behaviour features such as the maximum number of reviews, first review ratio, review length, rating deviation and maximum content similarity to the support vector machine classifier based on review dimension features and improved the recognition accuracy from 66% to 68% to 85%–86%. Literature (Zhang et al., 2016) verified that non-verbal features such as social behaviour features and metadata features are more important than language features. In particular, metadata contains rich temporal and spatial attributes. Santosh et al. (Kc and Mukherjee, 2016) extract temporal features through time series analysis of real and false reviews, and the classification model constructed by combining behaviour and text features has an accuracy of 89%–90%.

Literature (Zhiwei Guo et al., 2020; Goswami et al., 2017; Yin et al., 2021; Wang et al., 2016; Sedighi et al., 2019) combines the textual representation features obtained based on deep learning models with the reviewer's behavioural features to improve the classification method's stability on a commercial review dataset. Ao Li (Li et al., 2019) used a convolutional neural network model to learn text representation features automatically and encoded behaviour features through an effective coding method. Combining the two types of features achieved an accuracy of 93%; Wang et al. (Wang et al., 2016), through the tensor decomposition method, the reviewer and product features are vectorised, combined with n-grams features, the SVM classifier is used to achieve an accuracy of 90%. Lee et al. (Rastogi and Mehrotra, 2017) constructed a reviewer-IP-review tripartite graph model using observational features such as review and review features and node labels based on the estimation of the adjacent relation of nodes in the graph model collaborative classification, and an accuracy rate of 82% was achieved on the dataset.

The classification model that combines the features of reviewers and relationship dimensions has achieved good results on commercial datasets and has better accuracy when applied to data sets labelled with rules (Lin et al., 2014). The classification model constructed by the review feature has good robustness. Although the comprehensive application of multiple types of features can effectively improve classification performance, the direct entry of many initial features into the model can easily lead to high-dimensional feature space problems (Heredia et al., 2017; Capuozzo et al., 2020). Some studies use singular value decomposition (Guangyu et al., 2010), and information gain (Heredia et al., 2017; Koven et al., 2014), chi-square test (Karami and Zhou, 2015; Crawford et al., 2016), signal-to-noise ratio (Crawford et al., 2016), and neural network (Goswami et al., 2017; Hema et al., 2020) and other feature selection methods to analyse the significance and correlation of the feature set, through Select false characterisation features and feature combinations to achieve the purpose of improving recognition performance and reducing the computational cost. In addition to the above-mentioned review-classification method based on supervised machine learning, Li et al. (Li et al., 2013) proposed a Bayes-based Topic Spam detection method that uses the probabilistic topic model Latent Dirichlet Allocation (Ya et al., 2020) and the bag-of-words feature to obtain the topic distribution of reviews and compares the reviews on false and real topics. The probability of the above predicts whether the

review is false or true, and the accuracy of this method on the crowdsourced annotation data set is as high as 95%. Viviani (Goswami et al., 2017) used multiple feature aggregation modes and a multidomain decision-making model to score the falsity of reviews based on review metadata and reviewer behaviour characteristics. The optimal aggregation operator combination was used in commercial annotation datasets, and the recognition accuracy achieved above is 82%.

In summary, the method's overall performance using annotated data is relatively high, and the classification model based on the feature of the review gradually increases the detection accuracy as the language level involved in the deep feature (word, sentence, document, semantics). However, the accuracy of this type of model on the business review dataset is low, and the classification performance can be improved by fusing the reviewer and relationship features into the model. In the review feature, using the text representation feature acquired based on the deep learning model has achieved relatively stable prediction performance in various review datasets and has good generalisation ability. Among the reviewers' behavioural features, the characteristics of sudden review and deviation of ratings have strong false representational power. Simultaneously, time series analysis and feature selection techniques are also effective means to improve classification performance. Although the method of using annotated data has a high recognition accuracy, there are doubts about the accuracy and applicability of the annotated dataset used by such methods, which greatly affects the credibility of the final result.

5.2. Semi-labelled Data-Based detection method

Some studies have proposed semi-supervised machine learning methods due to the lack of a large number of accurately labelled datasets. That uses semi-labelled data to identify fake reviews and use classifiers to automatically label a large amount of unlabelled data to assist a small amount of labelled data for learning. Existing research is commonly used; the algorithm includes Co-training and Positive-Unlabelled learning.

5.2.1. Classifier based on Co-training

The Co-Training learning method (an algorithm given below) requires that the data have two different feature subsets that are sufficiently redundant and satisfy conditional independence. "Sufficient" means that each view contains enough information to produce the optimal learner. At this time, for any of these views, the other view is "Redundant", these two view conditions are independent for category tags.

Algorithm: 1 Co-Training Algorithm (Blum & Mitchell 99)

Given : Labelled Data L

Unlabelled Data U

Loop :

Steps 1. Train g₁ (Hyperlink Classifier) using L

2. Train g₂ (age Classifier) using L

3. Allow g₁ to label p positive and n negative examples from U

4. Move these self-labelled examples to L (beginning in Loop's first line)

Lee et al. (Li et al., 2011) are confident that the fake spammer continues to write a fake review, and the review feature reviewer behavioural feature can identify this. The literature (Li et al., 2011) uses two feature sets related to reviews and critically related. Based on a small number of labelled samples, an F-measure value of 69% is obtained on the testing dataset. The classification results are better

than fully supervised classification using small datasets. The CoFea (Zhang et al., 2016) and CoSpa (Zhang et al., 2016) methods proposed by Zhang et al. Use the comments terminology and syntax features to construct double views, and the recognition accuracy on the “Gold Standard” dataset is more than 80%.

5.2.2. Classifier based on Positive-Unlabelled

The Positive-Unlabelled learning method trains a binary classifier through labelled positive samples P and a large number of unlabelled samples U . The size of P in the Positive-Unlabelled problem is usually quite small, and it is difficult to expand the P set. In contrast, the size of the U set is usually large and easy to obtain. This feature closely matches the problem of false Review detection. The false review can be regarded as the P set, and unlabelled reviews are treated as the U set. Hernandez-Fusilier (Fusilier et al., 2013; Fusilier et al., 2015) took all unlabelled data as negative and selected 20 % of reviews from the “gold standard” dataset as positive, combined with the NB classifier to obtain F-measure is 84%, without using positive unlabelled learned classification F-measure value is only 58%. Li et al. (Li et al., 2014) believed that treating all unlabelled data as negative examples is not accurate. They used the Spy method to inject some “spy” positive samples into the U set to obtain a more reliable set of negative examples, significantly improving the model recall rate. The MPIPUL method proposed by Ren Yafeng (Ren et al., 2014) focuses on identifying samples that are easily misclassified into unlabelled samples. The model uses 20% of the “gold standard” data to obtain an accuracy of 84%. The classification method based on semi-supervised machine learning can achieve better classification performance with a small amount of labelled data and solve the problem of lack of labelled data to a certain extent. The Positive-Unlabelled learning algorithm has a more stable performance on different datasets and has better generalisation. However, in general, there are not many research studies using semi-supervised identification methods, and the use of false features is relatively simple, mainly for review dimension features.

Algorithm: 2 PU-Learning (Bing Liu 2003)

Input Data:	X_Data = the training set Y_data = target var (1 for the positives and not-1 for the rest)
Output Data:	$Predicted_y$ # fit the classifier and estimate $P(s = 1 y = 1)$ Classifier, $ps_1y_1 = \text{fit_PPU_estimator}(x_data, y_data, 0.2, \text{Estimator}())$ # estimate the probability that x_data is labelled $P(s = 1 x)$ $Predicted_s = \text{classifier.predict_proba}(x_data)$ # estimate the actual probabilities that x is positive by calculating $P(s = 1 x)/P(s = 1 y = 1)$ $Predicted_y = \text{estimated}_s/ps_1y_1$

5.3. Unlabelled Data-Based detection method

In order to overcome the problem of small amounts of labelled data and difficulty in determining the accuracy, some research studies have proposed detection methods that do not rely on labelled data.

5.3.1. Semantic analysis

Semantic analysis is a method of semantic information analysis based on natural language. It performs grammatical analysis such

as lexical analysis and syntactic analysis and involves the meaning contained in words, phrases, sentences, and paragraphs. The purpose is to use sentences. The semantic structure represents the structure of the language. Semantic analysis is divided into lexical-level semantic analysis and sentence-level semantic analysis, and a general semantic analysis model has shown in Fig. 13 (Salloum et al., 2020). For different language units, the tasks of semantic analysis are different. At the word level, the basic task of semantic analysis is word sense disambiguation, semantic role labelling at the sentence level, and referential disambiguation at the text level, also known as co-referential resolution.

1 Lexical-level Semantic Analysis:- The content of lexical-level semantic analysis is mainly divided into two parts:

(a) Word Sense Disambiguation:- The phenomenon that a word has multiple meanings in natural language is very common. How to automatically learn the multiple meanings of a word or confirm the meaning according to the context when a word is known to have multiple meanings is the content of word sense disambiguation research. The word bank may mean a bank or river-bank. Therefore, based on this current situation, the task of word sense disambiguation is to determine the meaning of the word according to the given input context. Semantic disambiguation methods are roughly divided into semantic disambiguation based on background knowledge, Supervised semantic disambiguation method, Semi-supervised learning method, and Unsupervised learning methods. The first method is a rule-based method (also called a dictionary method), and the latter three are machine learning methods. Semantic disambiguation methods are roughly divided into semantic disambiguation based on background knowledge, Supervised semantic disambiguation method, Semi-supervised learning method, and Unsupervised learning methods. The first method is a rule-based method (also called a dictionary method), and the latter three are machine learning methods (Chen et al., 2009).

Semantic disambiguation based on background knowledge:- The basic idea of the semantic disambiguation method based on background knowledge is as follows: the definition of the entry itself in the dictionary is used as the condition for judging its semantics. For example, the word apple has two definitions in the dictionary: one refers to “an apple is a fruit”, and the other refers to “an apple used to hold other things, such as an apple is a company”. If the text “fruit” or “company” and apple appear in the same context, then the semantics of apple can be determined, fruit corresponds to apple semantic 1, company corresponds to apple semantic 2. It can be seen that this method is completely rule-based (Chen et al., 2009).

Supervised semantic disambiguation method:- The method of supervised learning is that the type of data is known before learning. In the problem of semantic disambiguation, all possible meanings of each word are known. The supervised semantic disambiguation method is to learn a classification model through a labelled corpus. Commonly used methods are the Word sense disambiguation method based on Bayesian classifier, Word sense disambiguation method based on maximum entropy, Disambiguation method based on mutual information (Chen et al., 2009).

Unsupervised semantic disambiguation method:- Strictly speaking, it is impossible to use a completely unsupervised disambiguation method for word sense tagging, because after all, word sense tagging needs to provide some descriptive information about semantic features. However, word sense recognition can be achieved using a completely unsupervised machine learning method. The key idea lies in context clustering. Calculating the similarity of context vectors appearing in polysemous words can realise context clustering, thus realising word sense differentiation (Chen et al., 2009).

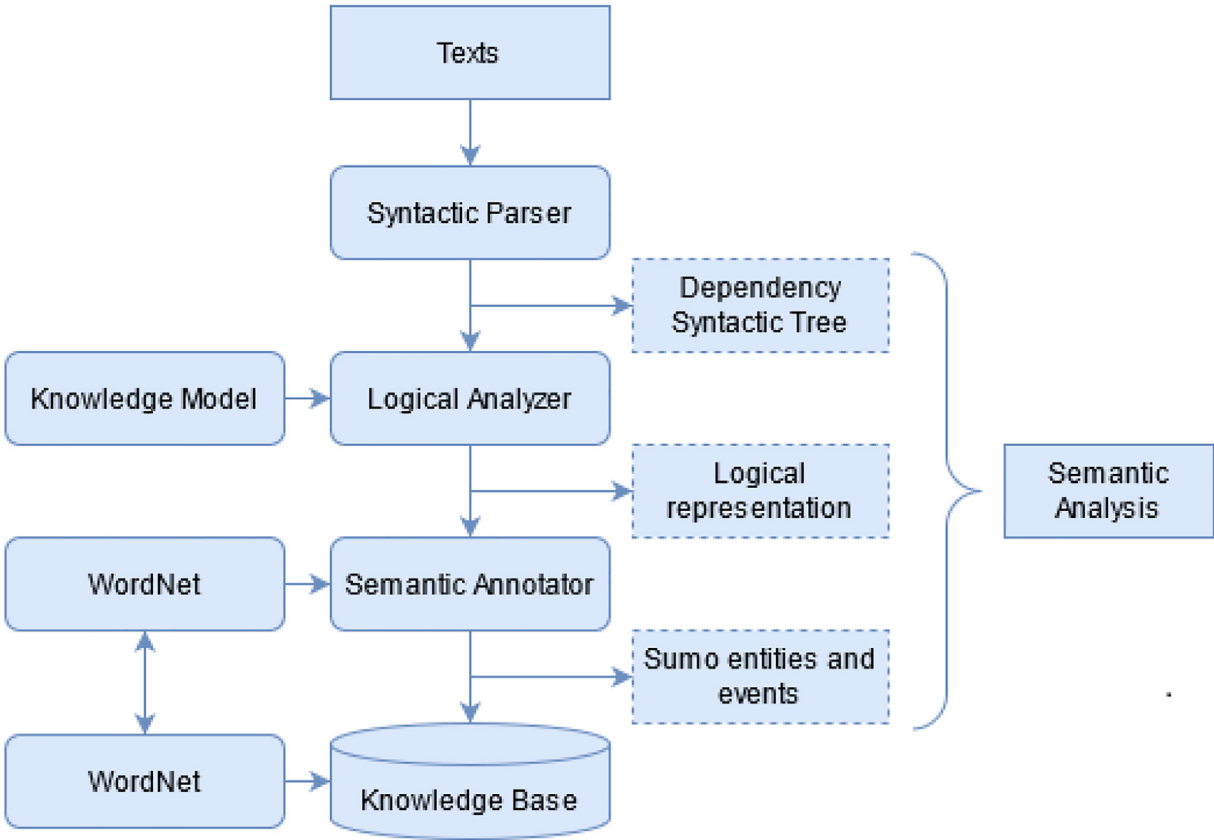


Fig. 13. A general model of semantic analysis.

(b) Text Similarity:- The meaning of text similarity is easy for everyone to understand. Here we use a complicated but slightly formal way to explain the meaning of word similarity: two words can be used interchangeably in different contexts without changing the syntax and semantics of the text the degree of structure. The greater the possibility that the text can be replaced in different contexts without changing the text's semantic structure, the higher the similarity between the two, otherwise the lower the similarity. A numerical value expresses the similarity, and the general value range is between [0,1]. The semantic similarity between a word and itself is 1; if two words are not replaceable in any context, the similarity is 0. It is worth noting that similarity involves vocabulary characteristics in terms of morphology, syntax, semantics and even pragmatics. Among them, the most important influence on word similarity should be word semantics. Word distance is another important index to measure the relationship between two words, represented by a real number between (0 to 1) (Mostafa et al., 2020).

2. Sentence-level Semantic Analysis: - Sentence-level semantic analysis is mainly divided into two parts: shallow semantic analysis and deep semantic analysis. Here we mainly introduce shallow semantic analysis.

Reference (Algur et al., 2010; Chuk Ling et al., 2010; Chuk Ling et al., 2010; Lau et al., 2012), built a semantic language model, calculated the similarity of comments at the semantic level, and artificially set the threshold of similarity to judge the falsity of the review; Reference (Kaur and Malik, 2021; You et al., 2020) used sentiment analysis technology (Saumya and Prakash Singh, 2018) to mine the emotional features in the review text. Quantify sentiment values (Sharma and Lin, 2013), sentiment outliers, and anal-

yse differences in sentiment scores (Zhiyuli et al., 2015; Algur and Biradar, 2015);(Saeed et al., 2020) to judge the falsity of reviews.

5.3.2. Clustering method

The literature (Li et al., 2016) transformed the review text classification into a clustering problem and used the K-means clustering algorithm (Jose and Sasidhar Babu, 2019; Chandra Pandey and Singh Rajpoot, 2019) to divide the review into two groups, false and non-false. Ren Yafeng et al. (Sun et al., 2016) used a genetic algorithm for feature optimisation and selected fuzzy k-means clustering to achieve an accuracy of 84.5 % on the “Gold Standard” dataset. Deng Song et al. (Zhang et al., 2016) combined reviews and reviewer behaviour characteristics and used K-means clustering based on the square error to achieve 90 % accuracy on manually labelled datasets.

Algorithm: 3 K-Mean Clustering Algorithm (James MacQueen 69)	
Given	: $D=\{t_1, t_2, t_3, \dots, T_n\}$ // Set of Elements
	K // No. of desired clusters
Output	: K // Set of clusters
Start	:
Repeat	Assign initial values for $m_1, m_2, m_3, \dots, m_k$ Train g_2 (Page Classifier) Using L Assign each item T_i to the clusters which have the closet means; Calculate the new mean for each cluster; Until Convergence criteria are met;

5.3.3. Time series method

Xie et al. (Xie et al., 2012) used a multi-time window method to detect false singleton comments by establishing a multi-dimensional time series. Hyderi et al. (Heydari et al., 2016) defined several reviews burst mode templates based on time series analysis, used pattern recognition technology to detect suspicious time windows matching the templates and combined the reviews and the reviewers to synthesise the comments' falseness in the time window. This method narrows the detection focus to several suspicious periods and reduces the computational cost of processing a large amount of text.

5.3.4. Markov Random field

There are two types of nodes in the Markov Random Field model (El-Mawass et al., 2020; Yongji et al., 2020; Wang et al., 2020), such as observation nodes and hidden nodes. The observation nodes contain some observable review or reviewer dimension features; each hidden node corresponds to an observation node, representing the true state of the data, that is, the label of the node. Literature (Rayana and Akoglu, 2015; Akoglu et al., 2013) constructs a network model, proposes the FraudEagle method (Akoglu et al., 2013) and SpEagle method (Rayana and Akoglu, 2015) based on MRF, and predicts the labels of hidden nodes through an improved back-propagation algorithm. The SpEagle method integrates the comment content and the reviewer's behaviour characteristics into the network model, and the recognition accuracy is significantly improved compared with the former. MRF denoted via graph $G = (V, E)$, $V = \{V_1, V_2, \dots, V_N\}$. Each node contains a random variable V_i , its neighbour is $N(V_i)$, and edge E denoted (V_i, V_j) . As per Markov Random Field above, node $N(V_i)$ and V_i does not depend on other nodes where node $N(V_i)$ is a Markov blanket.

5.3.5. Rule-Based detection method

Feng et al. (Feng et al., 2012) believed that extreme review and sudden review behaviours of spammers would lead to abnormal changes in the distribution of product ratings and the number of reviews. By analysing the distribution of abnormalities, they pro-

posed a rule-based false review detection method, which does not rely on review text having good versatility.

In summary, although this type of recognition method does not rely on labelled data for implementation, the verification of the method is still mainly based on unreliable labelled data, and the accuracy of the method remains questionable.

6. Discussion and analysis

The classification method based on supervised machine learning is the main method used in existing research. The application of the classifier is mainly a single classifier. SVM (Mewada et al., 2010) is the most used classifier. The performance of different classifiers in different sizes and different fields of datasets There is a big difference. Ensemble learning technology can improve the classification model's generalisation and robustness. Commonly used algorithms include Boosting, Bagging and Random Forests. Simultaneously, ensemble learning technology can deal with imbalanced dataset problems in machine learning (Algur et al., 2010; Chuk Ling et al., 2010; Chuk Ling et al., 2010). The classification method, the word embedding and the deep learning model are combined to automatically learn the review text feature to assist the feature design. The information loss caused by traditional artificial feature design can also reduce labelled dataset sparsity problems (Ren and Zhang, 2016). It shows a good recognition effect on a large data scale, especially on commercial data sets; the classification method's accuracy is easily affected. For the impact of the quantity and quality of training data, semi-supervised machine learning algorithms such as Co-training and positive unlabelled learning can be used. The classification performance can be improved by optimising labelled data (Ren et al., 2015). Existing unsupervised schemes, based on the Markov Random Field model, the network structure formed by the relationship between the reviewer, the review and the products have the strong fitting ability. However, the fusion of the dimensional features of reviews and reviewers is not enough, and the recognition performance needs to be further improved. The method based on time time-series analysis concentrates on identifying false targets in a few short periods, improving the method's accuracy and time performance.

Table 5

Summary and comparison of false review detection methods.

Method	Feature	Dataset	Outcome Matrix	Result
SVM (Ahmed et al., 2018)	N-gram	Hotels reviews	Accuracy	90%
Adaboost (Barbados et al., 2019)	Polarity and LDA	Yelp reviews	F-Measure	81%
DFNN and CNN (Barushka and Hajek, 2019)	N-gram and skip Gram	Hotel, Restaurant, Doctor	Accuracy	89%
k-NN and SVM (Elmurngi and Gherbi, 2017)	Unigram	Movie reviews	Accuracy	82%
Logistic Regression (Jindal and Liu, 2007)	Polarity, review Feature	Amazon reviews	AUC	78%
BERT (Kennedy et al., 2020)	Text feature, POS, Skip-gram	Yelp reviews	Accuracy	89%
Naive Bayes, Co-Training (Li et al., 2011)	N-gram, Polarity	Epinions	F-measure	63%
SAGE (Li et al., 2014)	Unigram, Polarity, POS Tag	Hotel, Restaurant, Doctor	Accuracy	65%
SVM (Li et al., 2015)	N-gram	Restaurant reviews	Accuracy	85%
CNN and SWNN (Li et al., 2017)	Sentence Level, POS Tag	Hotel, Restaurant, Doctor	Accuracy	84%
Logistic Regression (Liu et al., 2019)	Polarity, Text reature, POS Tag	Dianping	F-measure	81%
SVM (Mukherjee et al., 2013)	N-gram	Yelp reviews	Accuracy	86%
SVM (Ott et al., 2013)	N-gram	Hotels reviews	Accuracy	86%
SSL (Rayana and Akoglu, 2015)	Text Feature	Yelp reviews	AUC	79%
CNN and GRNN (Ren and Ji, 2017)	Continuous Bag of Words	Hotel, Restaurant, Doctor	Accuracy	84%
k-NN and Random Forest (Rout et al., 2018)	LIWC, Bigram and POS Tag	Hotels reviews	Accuracy	77%
SVM (Shojaee et al., 2013)	Text feature	Hotels Reviews	F-measure	84%
Bagging (Sun et al., 2016)	Word embedding, N-gram	Hotel, Restaurant, Doctor	F-measure	77%
SSL (Merve Yilmaz and Onur Durahim, 2018)	Continuous Bag of Words	Yelp reviews	Accuracy	83%
LSTM (Zeng et al., 2019)	Sentence 3 Partition	Hotel, Restaurant, Doctor	Accuracy	83%
SRD-BM and SRD-LM (Hussain et al., 2020)	Linguistic, Behavioural feature	Amazon Reviews	Accuracy	93%
One Class SVM (Tian et al., 2020)	Text feature, Ramp function	The gold standard by Ott	F-measure	93%
MGSD (Noekhah et al., 2020)	Review and reviewer Feature	Amazon reviews	Accuracy	93%

Table 5 summarises the main existing methods for spam review detection and comparing them from feature types and evaluation matrix. Existing research on methods for identifying spam reviews has achieved rich results. However, there are still some problems in spam review dataset acquisition, feature design and method design.

6.1. Future directions

1. Data Acquisition and Labelling Issues:- The primary issue related to the review dataset in the current research is the acquisition of a large-scale accurately labelled review dataset. Although the synthetic dataset is vast, the real and false reviews mixed, and the existing data labelling methods have many shortcomings, and there are particular misjudgement examples. However, crowd-sourcing's labelled data is more accurate than the distribution of real and fake review data is different from the reviews in the real world, which affects the accuracy of the classification results. Therefore, the acquisition of labelled data and the study of labelling methods are painful points in this field and also a problem worthy of research.

2. Feature Design Issues:- The existing research mainly relies on field experts' observation and experience, subjective and information omissions regarding feature design methods. The text representation features acquired based on the deep learning model can automatically learn the data's implicit rules and vectorise the text or structure. It has achieved high accuracy on multiple types of datasets, and its versatility is better than that based on manual. The design characteristics are worthy of more application and research; in the way of feature application, the emergent characteristics and timing characteristics of the review contain the characteristics and rules of the review behaviour in the time series. They have strong false representation capabilities, but these features use in the research are still rare. Besides, the fusion of multiple types of features and the selection and verification of effective features are areas where current research still has insufficient attention; in terms of feature data sources, existing features mainly come from analysing the review text and metadata. With the development of shopping website functions and services, there are more and more data sources related to comments, such as additional comments, user Question and Answering platform data, user social platform data, logistics information, how to use the new data sources reasonably, and Digging is also a direction worth exploring.

3. Method design issues

(a) Domain Adoption Issue:- Most of the existing research trains and tests models and methods in a single review field. A few studies (Li et al., 2014; Hernandez-Castaneda et al., 2017) have discussed and practised domain-independent development recognition methods. The research on domain migration is of practical significance. Different review domains have different capabilities to characterise falseness. Because it is very difficult to label data, it is impossible to label all review domains in practical applications. Therefore, applying the features and models obtained in one field to multiple other target fields is an important research direction.

(b) Data Imbalance:- To obtain the best classification results, most of the existing studies have constructed a labelled dataset to train a classifier. The ratio between the number of false reviews and the total number of reviews in the real world remains very low (Najada and Zhu, 2014). Researching from the perspective of reducing the impact of the imbalanced dataset on classification models is also an effective way to improve classification methods' performance.

(c) Big Data Issue:- The data volume of product reviews is huge, and many new reviews are continuously generated on different

platforms every day. It has typical big data (4 V) characteristics. The problem of false reviews detection is also a big data problem (Crawford et al., 2015; Kauffmann et al., 2020), so big data methods are applied in this field. The research is of great significance. Existing research mostly uses traditional machine learning tools, such as R language, Weka, which are not suitable for large-scale dynamic data processing. Such tools as Spark and SAMOA can perform large-scale online real-time learning have not yet been applied.

(d) Cold-start Issue:- Research shows that reviewers behavioural features. If the falseness of a newly published review is judged in real-time, a "cold start" occur due to lack of historical data (Algur and Biradar, 2015). Only a few studies (Kumar et al., 1703; Wang et al., 2017; Xue et al., 2015) discuss this issue worthy of further research.

7. Conclusion

With the rapid growth of the number of false reviews, identifying false reviews has attracted continuous attention from the academic community. From the perspective of the technical route of spam review detection research, this article classifies and summarises the key issues of the current false review text detection research, including the introduction and comparison of false feature design, recognition models and methods, and then summarises and discusses the existing problems and issues in current research. The future research direction references further research in this field regarding research directions, ideas, and methods. Since the literature reviewed in this article only covers research that uses false review text as the identification target, in the future, we can continue to sort out, summarise and compare the research that uses false reviewers and false reviewer groups as the identification target and further improve the overall research on false review identification—presenting the status quo.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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